

Multiple Choice (10 marks)

1. What is the limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule?
 - (a) $1.5 R$
 - (b) $2 R$
 - (c) $2.5 R$
 - (d) $3.5 R$
2. Which of the following is a primary standard for use in standardizing bases?
 - (a) Ammonium hydroxide
 - (b) Sulfuric acid
 - (c) Acetic acid
 - (d) Potassium hydrogen phthalate
3. A molecule has a rotational correlation time of 1 ns. At what magnetic field would protons in this molecule have the fastest spin-lattice relaxation? [Use eqns 5.3 and 5.4.]
 - (a) 3.74 T
 - (b) 5.19 T
 - (c) 6.08 T
 - (d) 9.49 T
4. Calculate the equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K.
 - (a) 10.8×10^{-5}
 - (a) 4.11×10^{-5}
 - (a) 3.43×10^{-5}
 - (a) 1.71×10^{-5}
5. Of the following compounds, which has the lowest melting point?
 - (a) HCl
 - (b) AgCl
 - (c) CaCl_2
 - (d) CCl_4

True / False (10 marks)

Answer True or False

6. True or False: The +3 oxidation state is more stable than the +1 oxidation state for thallium (Tl) because of its higher electronegativity.
7. True or False: The thermal stability of group-14 hydrides increases in the order: $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$.
8. True or False: Silicon naturally occurs most frequently as a metallic element in the Earth's crust.
9. True or False: 0.050 M AlCl_3 has the highest ionic strength due to its complete dissociation into four ions per formula unit.
10. True or False: Planck's quantum theory is primarily used to explain the emission spectra of diatomic molecules in detail.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reason why the ^1H NMR spectrum of CHCl_3 appears as a singlet, focusing on the role of chlorine isotopes and their electric quadrupole moments.
12. Calculate the magnetogyric ratio for ^{23}Na given its NMR frequency of 198.52 MHz in a 750 MHz spectrometer. Discuss the significance of this ratio in NMR spectroscopy.

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